MICHAEL FERNANDES

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EDUCATION

M.Sc. in Big Data Analytics June 2025 – April 2027
St. Xavier's College, Mumbai **CGPA: 9.4 / 10.00**

B.Sc. in Information Technology June 2022 – May 2025
Sahyog College of Management Studies, Thane (W) **CGPA: 9.2 / 10.00**

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, Unix/Linux Scripting
Data Science : Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, , Streamlit, Plotly, Matplotlib
Quantitative Finance: QuantLib, Backtrader, TA-Lib, Statsmodels, gs-quant
Cloud & Databases: AWS, MySQL, PostgreSQL, MongoDB, kdb+/q, DuckDB
Data Engineering: Apache Spark, Kafka, Airflow, Hadoop, Iceberg, Snowflake, Databricks, dbt,

EXPERIENCE

Data Analyst **September 2023 – Present**
Asterix StratComm — Mumbai, India

- Designed and optimized scalable Python ETL pipelines using Pandas and NumPy to ingest, clean, transform, and standardize 80K+ FMCG records, reducing data processing time by 65% while improving pipeline efficiency.
- Built automated data quality and preprocessing frameworks incorporating schema validation, missing-value imputation, duplicate detection, and consistency checks, increasing data accuracy by 30% and eliminating repetitive manual validation.
- Developed interactive Streamlit analytics dashboards featuring KPI tracking, advanced filtering, and drill-down capabilities, enabling business stakeholders to generate real-time market insights and reducing reporting turnaround by 70%.
- Engineered reusable data transformation workflows and modular processing components, improving code maintainability, accelerating feature delivery by 50%, and supporting scalable analytics across multiple datasets.
- Performed advanced Exploratory Data Analysis (EDA) and statistical profiling to identify pricing trends, product portfolio, product segmentation, and regional patterns, driving data-informed strategic decisions.
- Enforced different analysis sections like Pricing Analysis, SKU Analysis, Brand and Variant Segmentation, Geographical segmentation across six Category Segments

PROJECTS

Quantitative Equity Research & Market Analytics Platform
Tech: Python, LSTM, GRU, RNN, TA-Lib, yfinance, Streamlit

- Built a modular quantitative research platform integrating feature engineering, technical analysis, and deep learning (LSTM, GRU, RNN) for equity price forecasting.
- Benchmarked forecasting models using RMSE and MAE to evaluate predictive performance across architectures.
- Designed a multi-factor stock screener combining financial and technical filters for candidate selection.
- Implemented technical indicators including RSI, MACD, and Bollinger Bands for trend and momentum analysis.
- Built interactive Streamlit dashboards for stock comparison, historical trend analysis, and model output visualization.

Multi-Factor Equity Portfolio Optimization Platform
Tech: Python, Scikit-learn, Pandas, NumPy, SciPy, CVXPY, GS-Quant, Plotly, Streamlit, Scikit-learn,yfinance

- Built an end-to-end quantitative portfolio optimization system using Python, GS-Quant, Pandas, NumPy, CVXPY, Streamlit, and Plotly.
- Implemented multi-factor models using Momentum, Value, Quality, and Volatility signals to rank and select equity investments.
- Developed constrained mean-variance optimization and monthly rebalancing strategies for systematic portfolio construction.
- Performed portfolio backtesting, performance attribution, and benchmark comparison using historical market data.
- Designed a risk management framework incorporating Historical VaR, CVaR, Beta, Maximum Drawdown, Sharpe Ratio, Sortino Ratio, and rolling risk analytics.
- Portfolio Optimization over NIFTY50 Stocks using yfinance price data